



ASSIGNMENT 1

SEPTEMBER 2025 SEMESTER

SUBJECT CODE	:	CCF201
SUBJECT TITLE	:	COMPUTER FORENSIC
LEVEL	:	BACHELOR

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INSTRUCTIONS TO STUDENTS

- 1) This assignment consists of **TWO (2)** questions. Answer **ALL** questions.
- 2) Plagiarism in all forms is forbidden. Students who submit plagiarised assignment will be penalised.
- 3) Your assignment will be examined based on the followings
 - a complete working solution.
 - ability of using methods available in the learning materials.
- 4) This assignment carries a **30%** weightage toward final grade.

THERE ARE TWO (2) PAGES OF QUESTIONS, INCLUDING THIS PAGE.

INSTRUCTION. Answer ALL questions. [Total : 30 Marks]

Question 1

Researchers have found that systemic weaknesses in traffic control systems make them vulnerable to be attacked by cybercriminal. Those weaknesses include the use of unencrypted wireless signals to control the lights, the use of default usernames and passwords on these control systems, and a vulnerable component known as a traffic controller, which is the device that controls the lights and walk signs.

Briefly explain the role of computer forensic and the **SEVEN (7)** steps taken in investigating cybercrime.

[10 Marks]

ANSWER:

Computer forensics plays a crucial role in investigating cybercrimes. Computer Forensics involves handling digital evidence in a scientific and legal manner. Computer Forensics aims to identify the activities in a crime scene involving digital devices and the individuals involved in the case while maintaining a documented chain of evidence that is applicable in a court of law. (Kaur, 2025)

In attacks on traffic control systems, computer forensics helps investigators:

- Identify how attackers exploited weaknesses such as unencrypted wireless signals or default passwords
- Trace unauthorized access to traffic controllers
- Recover deleted, hidden, or altered data
- Determine who was responsible, what was affected, and the extent of damage
- Ensure evidence is admissible in court by maintaining integrity and chain of custody

Overall, computer forensics helps in incident response, legal prosecution, system recovery, and prevention of future attacks.

The seven steps that are taken while investigating Cybercrime are:

1. Identification

- The first step in investigating a cyber crime is identifying the crime itself. The process begins by determining what has occurred, identifying any digital evidence that may be present in the crime scene such as computers, mobile phones, storage devices, cameras, etc. (BimStudies, 2025)
- For example, traffic lights changing abnormally or pedestrian signals malfunctioning may indicate a cyberattack. Investigators identify affected systems, such as traffic controllers or wireless networks.

2. Preservation

- After identifying the suspect devices, they are seized and sealed off to prevent manipulation of data/evidence. The devices are registered and stored in a tamper-proof environment and sealed to everyone but the investigation team in order to maintain the integrity of the evidence. The devices may be imaged to create an exact replica of the system including the hash of the system for investigation without tampering with original evidence and to maintain integrity. (Acharya, 2025)
- For example, this includes isolating the compromised traffic control system to prevent further damage and ensuring that logs and data are not altered.

3. Extraction

- In this step, the data is extracted from the suspect devices. If the devices are on in the crime scene, then volatile data such as RAM is recovered using specialized techniques and devices in order to capture running processes and temporary data. In the investigation of storage devices, the images are investigated and their data is extracted instead of the original device. (Web Asha Technologies, 2025)
- For example, this may include system logs, wireless communication data, configuration files, memory dumps, and controller firmware.

4. Examination

- In this step, the collected data is examined to locate relevant evidence. Investigators filter out unrelated files and focus on suspicious activities such as unauthorized logins, abnormal commands sent to traffic lights, or configuration changes. (Web Asha Technologies, 2025)

5. Analysis

- Analysis involves interpreting the examined data to reconstruct the cyberattack identify evidence or proof that suggests misconduct or suspicious activity. This involves recovering and inspecting erased, corrupted, or encrypted file. Durin the analysis, investigators determine how the device was used, who used it, timeline of events, etc, which may be viable evidence in court. (BimStudies, 2025)
- For example, investigators determine how the attacker gained access, which vulnerabilities were exploited, the timeline of events, the impact on traffic control operations, etc.

6. Documentation

- Every activity during the investigation is documented. This includes tools used, procedures followed, findings, and evidence handling steps. Proper documentation of investigation ensures transparency and supports legal proceedings. (Web Asha Technologies, 2025)

7. Presentation

- Finally, the findings are presented/submitted in a clear and structured report in the court that will decide how to proceed with the handing of the case. (BimStudies, 2025)
- The report explains technical details in simple terms for law enforcement, management, and courts.

Question 2

Digital forensic investigations are commonly performed to explore hard drive content. This is where a dead analysis comes into the equation.

- a) With an appropriate example, describe “dead analysis” in digital forensics.

[10 Marks]

Answer:

Dead analysis in Digital Forensics is the process of collecting and examining digital evidence after a computer or device has been powered off. The focus of the investigation is the capturing of non-volatile storage media such as hard disks, SSDs, USB drives, or

memory cards in favor of volatile memory such as RAM, register, and cache data. The data recovered from dead analysis are Full forensic image of the hard drive, system and user file, (system, network and event) logs, configuration files and stored credentials, etc. (Lark Editorial Team, 2024)

Dead analysis is performed when there is high risk of data tampering when extracting live evidence or suspicion of evidence destruction through network connections or timed script execution. This method ensures that no new data is written to the device during examination, preserving the original state of evidence. (4Geeks, 2025)

Dead Analysis is performed by following the following steps:

1. The suspect system is safely shut down.
2. Storage media is removed or accessed externally.
3. A forensic image (bit-by-bit copy) is created.
4. Write blockers are used to prevent modifications.
5. Analysis is performed on the image.

For example, in a financial fraud investigation, an accountant is suspected of manipulating accounting records. Investigators power off the computer and perform dead analysis on the hard drive. They recover deleted spreadsheets, examine access timestamps, and find unauthorized file modifications, evidence that can be presented in court.

The advantages of dead analysis are:

1. There is Minimal risk of evidence tampering.
2. It provides strong legal admissibility.
3. It is easier to document and has high repeatability.
4. It is suitable for analyzing detailed file systems. (4Geeks, 2025)

The limitations of dead analysis are:

1. Volatile data such as RAM, cache that may contain running processes and temporary files are lost. (Lark Editorial Team, 2024)
2. Investigators cannot detect and examine live network connections. (4Geeks, 2025)
3. It is difficult to access encrypted disks after they have been powered off. (4Geeks, 2025)

- b) Traditionally, computer forensics experts agreed that shutting the computer system down in order to preserve evidence and eliminate the potential changing of information is best practice prior to examination. The evolution of technology has put us face to face with the harsh reality that it is sometimes more advantageous to perform “Live” analysis than the traditional one. Discuss why the “live analysis” is needed in digital forensics.

[10 Marks]

Answer:

Live analysis refers to the collection and examination of digital evidence while the system is powered on and operational. This approach allows investigators to capture volatile and real-time evidence that would be lost when the device is powered down (BlueVoyant, 2025).

Why Live Analysis Has Become Necessary

1. Recovery of volatile data

- Volatile data stored in RAM disappears when the system shuts down. The data that is lost when the system powers down are running processes, open files, active network sessions, logged-in users, etc. (FasterCapital, 2025)
- Live analysis allows investigators to preserve this critical information. Accessing running processes helps in understanding the exact situation of when the crime may have occurred, so preserving this information is very crucial. (FasterCapital, 2025)

2. Encryption challenges

- Modern systems are encrypted at full-disk and file-level through technologies like hardware bit locker. (4Geeks, 2025)
- If the system is powered off, encryption keys stored in memory are lost, making data inaccessible. Though live analysis, investigators can analyze the decrypted data. (4Geeks, 2025)

3. Fileless and Memory-Resident Malware

- There are malwares that run entirely in memory, delete itself on shutdown, or avoid writing files to disk. Live analysis is often the only way to detect such threats. (FasterCapital, 2025)

4. Detect ongoing cyberattacks

- Running Live analysis on suspect hardware may be able to detect ongoing threats/ cyberattacks that may be lost when the system is powered off.

5. Real-time network evidence

- Live analysis can capture command-and-control connections, data exfiltration activities, remote attacker IP addresses. This evidence is essential for tracing cybercriminals. (FasterCapital, 2025)

6. Cloud and virtual environments

- Modern computing relies heavily on cloud computing, virtual machines, and containers, these environments require live analysis to capture session data and running instances. (FasterCapital, 2025)

The risks and challenges of live analysis

1. There is a higher risk of altering evidence when investigating a live machine due to lack of imaging and hashing to provide integrity of the system. (4Geeks, 2025)
2. It requires skilled forensic professionals in order to prevent accidental alteration of evidence.
3. It requires following strict documentation and legal procedures. (4Geeks, 2025)

[Total : 20 Marks]

END OF ASSIGNMENT QUESTIONS

References

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